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CS 499 Professional Self-Assessment

I am a recent graduate majoring in computer science. After completing all required professional training projects for my program and building my dedicated CS 499 electronic portfolio, I have clarified my personal strengths, career goals, and values. Over the course of my university studies, I acquired seven core technical competencies, growing from only being able to write basic programs to someone who can master the full-process logic of software development. This portfolio integrates all of these abilities, and it supports my application for technical roles in the technology industry.

One of the core competencies I have focused on is software engineering and design. I first acquired skills including code organization, separation of responsibilities, and maintainability improvement in the CS360 course. I later revised the `ValidationUtils.java` and `InventoryActivity.java` files for the inventory application in my personal ePortfolio. Ultimately, I gained a clear understanding that professional software must possess the core attributes of readability, consistent structure, reusability, and ease of maintenance.

I organized the practical growth and cognitive advancement I achieved through this project along two parallel technical dimensions. The first dimension covers the field of algorithms and data structures: I modified an enhanced inventory application, added the `InventoryItem.java` class, used `ArrayList` to manage inventory, and launched the functions of search, sorting, filtering, low-stock display, and full inventory display. Through this process, I gained firsthand understanding that algorithms and data structures can effectively improve the application's information processing efficiency.

The second dimension focuses on database development: I modified the DatabaseHelper.java file, and added functions including table constraints, duplicate entry checks, secure CRUD operations, password hashing, and data validation. This experience led me to recognize that databases fulfill the core roles of storing data, protecting data, and boosting system reliability. This insight aligns with my interest in cybersecurity, and has anchored the core value of secure data processing.

Various projects completed as part of my computer science major helped me build my communication and collaboration skills. Coursework required me to explain technical decisions using documents and charts, while my professional work experience demanded that I coordinate with a range of stakeholders including team members and users. Materials stored in my electronic portfolio, such as code review videos, provide evidence to verify this competency.

Security has become one of my strongest career interests. Through secure coding, database work, and my interest in cybersecurity and data center technology, I learned that security should be considered throughout the software development process. My enhancements show this mindset through input validation, safer data handling, password hashing, and permission awareness. These skills support my career goals in software development, networking, data center operations, and cybersecurity.

My job-focused electronic portfolio for computer science covers the six core domains of the CS major. It strings together cross-domain practical content through the course project CS 360 Inventory Management Application, from which five hands-on project execution skills are distilled. These skills support my career growth as a CS graduate, and qualify me for a wide range of career paths including software development, data center roles, networking, and cybersecurity.